# AWS S3 Practical Lab Guide

## Task 1: Create Your First S3 Bucket

Log in to AWS, open Amazon S3, click Create bucket, provide a globally unique bucket name, choose a Region, keep default settings, and create the bucket. Verify that the bucket appears in the bucket list.

## Task 2: Upload and Download Files

Upload a text file, image, and PDF to the bucket. Download each file and verify it opens correctly. This demonstrates how S3 stores objects.

## Task 3: Organize Objects

Create folders such as assignments, images, documents, and backups. Upload files into each folder to understand prefixes and logical organization.

## Task 4: Object Management

Copy a file, move it to another folder, delete an object, and understand that renaming is achieved by copying to a new name and deleting the old object.

## Task 5: Versioning

Enable bucket versioning. Upload a file, modify it, and upload it again. View all versions and restore an older version after deletion.

## Task 6: Permissions

Review Block Public Access, Bucket Policy, and ACLs. Explain why S3 buckets are private by default and when public access should be used.

## Task 7: Static Website

Enable Static Website Hosting, upload an index.html page, configure read access, and browse to the generated website endpoint.

## Task 8: Pre-Signed URLs

Generate temporary download links that expire after 5 minutes and 1 hour. Test access before and after expiry.

## Task 9: Storage Classes

Upload files using Standard, Intelligent-Tiering, Standard-IA, One Zone-IA, and Glacier Instant Retrieval. Compare use cases.

## Task 10: Lifecycle Rules

Create a lifecycle policy to transition objects to Standard-IA after 30 days, Glacier after 60 days, and delete after 365 days.

## Task 11: Metadata

Upload a file with custom metadata such as Department=DevOps and Batch=AWS-01, then verify it.

## Task 12: Encryption

Upload files using SSE-S3, SSE-KMS, and no encryption. Compare the differences and security benefits.

## Task 13: Access Logging

Enable server access logging and store logs in another bucket. Access objects and verify that log files are generated.

## Task 14: Cross-Region Replication

Create buckets in two regions, configure replication, upload files to the source bucket, and verify they appear in the destination bucket.

## Task 15: AWS CLI

Configure the AWS CLI, create a bucket, upload/download files, list objects, delete objects, and remove the bucket using CLI commands.

## Mini Project

Create a student backup solution with folders, versioning, default encryption, lifecycle rules, pre-signed URL, CLI upload, and screenshots documenting each step.